

EMIPredict AI - Intelligent Financial Risk Assessment Platform

Machine Learning Project Report

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1. Problem Statement

Build a comprehensive financial risk assessment platform that integrates machine learning models with MLflow experiment tracking to create an interactive web application for EMI prediction. Nowadays, people struggle to pay EMI due to poor financial planning and inadequate risk assessment. This project aims to solve this critical issue by providing data-driven insights for better loan decisions.

The project specification calls for classification of EMI eligibility, regression of maximum EMI amount, real-time financial risk assessment, feature engineering, MLflow integration, and Streamlit Cloud deployment.

2. Objective

This implementation focuses on predicting EMI eligibility as Eligible, High_Risk, or Not_Eligible using customer demographic, employment, financial, credit, and loan-application information.

3. Dataset

The working dataset contains 404,800 records and 27 columns. The target variable is **emi_eligibility**. The project specification describes 22 input variables and two target variables in the broader platform: EMI eligibility classification and maximum monthly EMI regression.

4. Data Preprocessing

Mixed data types were converted, missing numerical values were imputed, categorical fields were retained for encoding, and duplicate and negative-value checks were performed. No duplicate rows or negative numerical values remained. Unusual credit-score values were identified and handled; the final credit-score range was 407 to 850 with no missing values.

Final cleaned dataset: **404,800 × 27**; total missing values: **0**; duplicate rows: **0**.

5. Exploratory Data Analysis

The EDA covered demographic distributions, employment and income, financial obligations, credit information, requested loan amount and tenure, and EMI eligibility.

Gender distribution was 60.02% Male and 39.98% Female. Marital status was 76.05% Married and 23.95% Single. Education was 45.31% Graduate, 24.78% Post Graduate, 15.00% High School, and 14.90% Professional.

Target distribution was 312,868 Not_Eligible, 74,444 Eligible, and 17,488 High_Risk. Eligible customers had higher mean monthly salary (■78,860.94) and credit score (725.17) than Not_Eligible customers (■54,257.63 and 693.65 respectively).

6. Feature Preparation

The classification matrix contained 24 input features. The data was split into 323,840 training records and 80,960 testing records. Eight categorical features were one-hot encoded and numerical features were scaled, producing 42 transformed features.

7. Models Evaluated

Logistic Regression, Random Forest, and Balanced Random Forest were trained and evaluated. XGBoost was considered but was not installed in the working environment and therefore was not trained.

Model	Accuracy	Macro F1	High_Risk Recall
Logistic Regression	78.67%	63.10%	65.48%
Random Forest	91.65%	61.20%	1.34%
Balanced Random Forest	91.73%	61.32%	1.34%

8. Logistic Regression Results

Logistic Regression achieved 78.67% accuracy, 64.91% macro precision, 73.19% macro recall, 63.10% macro F1, 65.48% High_Risk recall, and 23.02% High_Risk F1. The multiclass ROC-AUC was 0.9237.

The confusion matrix showed 10,934 Eligible, 2,290 High_Risk, and 50,464 Not_Eligible test cases correctly classified.

9. Tuning and Feature Importance

Hyperparameter tuning for Logistic Regression used 3-fold cross-validation. The best parameter was C=1 with cross-validation macro F1 of 0.6294. Important features included requested_amount, requested_tenure, house_type_Rented, college_fees, existing_loans_Yes, school_fees, groceries_utilities, current_emi_amount, bank_balance, and monthly_salary.

10. New Customer Prediction

A new customer was passed through the saved preprocessing pipeline and Logistic Regression model. The prediction was **Not_Eligible**, with the reported probability 1.0000000000 for that particular input. This is a case-specific model probability, not overall model accuracy.

11. Model Saving

The final Logistic Regression model and preprocessing pipeline were saved successfully for reuse and future predictions.

12. Key Findings

The target is imbalanced. Random Forest and Balanced Random Forest produced higher overall accuracy, but both had only 1.34% High_Risk recall. Logistic Regression had lower accuracy but substantially better High_Risk recall (65.48%) and strong multiclass ROC-AUC (0.9237). Therefore, accuracy alone should not determine model suitability for risk-sensitive applications.

13. Limitations and Future Scope

The source specification also requires a regression solution for maximum EMI, at least three regression models, MLflow experiment tracking and model registry, and a Streamlit Cloud application. These components were not completed in the coding workflow documented here. Future work can add these components and further address class imbalance and High_Risk detection.

14. Conclusion

EMIPredict AI demonstrates an end-to-end classification workflow for EMI eligibility assessment. The implementation successfully processed 404,800 records, completed preprocessing and EDA, trained and compared multiple models, evaluated class-specific performance, generated a new-customer prediction, and saved the model pipeline. The results emphasize the importance of class-specific metrics when building financial risk assessment systems.